

Active InferAnt Stream 005.1 ~ FORMINDEX: FORMIS INtegrated Database

ActiveInferAntStream | FORMINDEX: FORMIS INtegrated Database EXploration | 1:03:02

All right. Hello and welcome everyone. This is ActiveInferAnt stream number 5.1 October 16th, 2024. We're going to begin with a GitHub push.

I'm going to check it's been updated in GitHub.

This is also published on Zenodo. So it has a DOI. It's citable. And to show how to get a DOI on Zenodo, I'm going to make release 0.3 right now, adding a new tag 0.3.

ActiveInferAnt stream number 5.1. Publish it.

Published on GitHub.

Come back to Zenodo.

And we can find it citable. It just takes a few minutes for it to archive it.

All right. So here we are.

I have some topics and sections to go through.

And then if you're watching live, please feel free to write any comments and questions in the live chat. And this should be pretty interesting.

There's going to be a lot of ant bibliographic methods, as well as some generative AI, synthetic intelligence methods, usage of perplexity, LLMs, and so on.

Okay. So we began with a welcome.

Pushed the live stream update to GitHub.

We'll give a brief overview of the form index project.

I'm using cursor version 0.42.3.

Made the new version and the repo is listed here.

So what is the form index project?

Here we are in the form index coda, which is linked in the video description.

So the form index is a multi-stage project related to ant literature.

So in ant literature, there's the formis or formi, depending on how you pronounce it.

And this was an effort, is an effort, that curated citations related to ants.

And it has completeness in taxonomic areas of ant studies.

So a lot of rare ant taxonomy literature complete up to 1996 and then not as much after.

And so in form index, I'm setting out on a dual journey to do an analysis of formis for its own internal sensemaking,

to understand formis as a literature resource itself,

and then also to look more broadly towards integration of knowledge methods for mirror ecology,

for ant research, and then eventually even go beyond that with meta and format

and integrating bioinformatics and so on.

So in this first part of the stream, it's going to begin with the formis analysis,

which is going to briefly move through the ant-specific parts
about how the bibtext bibliography with 80,000 references is brought in,
and then some of the visualizations that come out of the just looking at formis itself.
Then the second part of the methods will focus on generative AI.
And we'll get there when we get there.
So first for the formis analysis.
So first, Sanford sent 316 sample citations.
And on October 8th, I read in those just to ensure I could get the format going.
He then sent over the bibtext plaintext file with 80,000 records,
which I brought in,
in the script section.
So first, I'll run this.
So first, reading in formis,
and visualize.
We'll run them.
So first, reading in formis,
we'll run this.
Read in formis.
This was from another computer, so we'll see.
Let's fix it on the go.
Meanwhile, though, continuing on the report.
This script reads into bibtext and stores the JSON format.
The generate target bibliographies subsets the 80,000 row JSON database
into bibliographies that mention that term.
So this is what it looks like for bioenergetics.
It's a JSON that's a strict subset of the overall.
We'll focus more on running the code for the second part
that I've more recently done on this computer.
The visualize formis.py.
Let's see if this one works.
Also, this one doesn't work,
but for the all formis database
and for every target bibliography,
some kinds of images are made.
So here's the distribution of records.
Here's histogram of record types.
So there's 60,000 articles,
11,000 tech reports,

5,000 collections,
3,500 PhD theses,
1,592 books,
one miscellaneous.

Here's
authors
by number of citations.
anon.

Coming through.

Coming through.

J.C.

DeLavis only slightly behind.

Publications per year.

So as mentioned earlier,
this is not the overall number of publications per year
in Google Scholar or literature database.

This is just for formis.

And this reflects what could be a really interesting pattern to continue exploring
as we integrate other literature searches.

We know that ant publications did continue after 1996,
and there's other citations.

So it's an interesting question.

How do we
add in?

What should be the scope of this database?

What keeps it ant relevant?

What's the relevance realization for the ant literature?

Here's a co-authorship heat map.

Some of these could be cleaned up
with

different visual improvements.

It's showing here, though,
the number of co-authorships between different authors.

So pretty cool to identify all these dyads.

And in the literature,
we could even look for higher order authorship combos.

Here's top venues.

So locations for publishing.

Insect social.

Evolution of the venues through time.

Locations.

Word cloud.

For the title.

Hymenoptera.

For Misedye.

And in the abstract.

Word cloud.

So those are the visualizations.

That's just to get pretty fast through the Formis method and get to the generative AI part where we're going to have active inference and all these other fun applications of AI.

So there's going to be three AI.

So there's going to be three AI, generative AI method sections here.

Notebook LM, OpenAI, API, and then perplexity.

And there's going to be the most to explore in perplexity.

So first, notebook LM.

Notebook LM.

Here, I took the target bibliography subsets.

And in the JSON format.

And in the JSON format, uploaded to a new notebook at notebook LM.

Made a podcast conversational style from those citations.

Then uploaded it to Revit AI to make these.

So I'll mute my local microphone and just let's see if the audio will come through just to hear what it's like for a second.

Ever looked at an anthill and thought, what's the story there?

Well, today, we're going full anthropologist on that.

I like it.

We're diving into the world of the red harvester ant.

Okay.

Or Puganomyrmex Barbados.

Okay.

Awesome.

So follow on through the links to hear the podcasts and really promising for local text-to-speech generation in the future.

Okay.

Second is the closest to what was previously done in some of the earlier inference streams where I used the OpenAI API to concatenate the target bibliographies with a general summarization prompt.

That's the pro summary, just the concatenated text file, kind of like the pre-pro grants from early on.

And then script two uses the OpenAI API to get the summary of the literature.

And then script three takes in the literature summaries and translates into target languages.

So this is more examples of how different literature resources, different text inputs can be summarized for different audiences and in different human languages.

Okay.

Now we're going to get to the fun and the new stuff with Perplexity AI.

Thank you, Tucker, for your message.

Making you a moderator for that one.

I hope you enjoy your new wrench, Tucker.

Okay.

So just to get familiar with what Perplexity is, here I pasted a prompt into Perplexity.ai, write a comprehensive summary of the rxinfer.jl package.

And Perplexity goes through a series of steps.

It does online searches, finds relevant chunks of information, and then it writes this report responding to the prompt with internet information, including these code boxes.

So I'm showing that because what this next section is going to do is essentially send this exact prompt, among others, and then interface with it through the scripts.

So here's another example.

Create a comprehensive grant proposal outlined for the NSF safety, security, and privacy of open source ecosystems, SafeOS program.

And then, again, similarly, it just thinks, okay, which ecosystem?

How about TensorFlow?

So enhancing security and privacy in the TensorFlow ecosystem, and then writes the grant, making up the information.

Also on the notebook LM, a friend just shared with me Illuminate, which you can do the literature search.

This is illuminate.google.com.

Do the literature search.

So here I had it look up active inference, and then had it pull in these four papers.

So it's taking one more step for you.

Whereas with notebook LM, you had to download the PDF and upload them.

Let's now listen for the first time to Illuminate.

I'm going to turn off my microphone, and it will play from the computer.

This conversation is powered by Google Illuminate.

Check out illuminate.google.com for more.

Welcome to our discussion on active inference, a fascinating framework for understanding how agents make decisions in uncertain environments.

We'll be exploring several research papers that delve into the intricacies of active inference, comparing it to the well-established field of reinforcement learning.

What are the core tenets of active inference, and how does it differ from reinforcement learning?

Active inference, rooted in the free energy principle, posits that agents act to minimize their surprise, the uncertainty about the world.

Unlike reinforcement learning, which focuses on maximizing a reward signal, active inference aims to minimize free energy,

a measure encompassing both prediction error and the complexity of the agent's internal model.

This...

Another...

Another one.

Pretty cool, though.

Illuminate.google.

Okay.

And Gen24 was with different kinds of calls to action translated into a lot of language, if people want to look in that repo.

Okay.

So, where were we with perplexity?

So.

Uh...

I did several things in the report on the 13th, and then in the several days since, several other things have happened.

So, I'm going to just skip over the earlier parts that were written about perplexity, and we'll pop over to the documentation for the API,

and then look at how it's being used in the package now.

So, where things are at right now, pending the rest of this stream, of course.

Communicating with the Formus stakeholders.

Get their feedback.

Get others' feedback.

Open Science, etc.

Then we have the analysis within Formi.

Looking at how we can do all kinds of augmented analyses within the database, as it stands, in July 2024 snapshot.

And then also, go beyond Formi, July 2024 snapshot.

So, adding in new literature, connecting with the species ID, perplexity searches, etc.

Fabric integration.

Kind of like what's happening in the previous InfraAnt stream.

So, okay.

All this is open source in the Formindex repo.

So, quick look at the perplexity API.

Very simple docs page.

You get \$5 a month for paying \$20 a month.

Wow.

And there's a really short, simple example that gets you going.

Here are the supported models they have at this time.

They have several of sizes of this perplexity sonar.

LAMA 3.1 base at 870 and 405 billion parameters.

They have several perplexity chat models.

These are chat.

They're not doing online searches.

And then there's the instruct open source models.

They mention the rate limit, 50 a minute.

We might hit that.

Pricing is shown here.

And each one of these requests seems to be costing using the large model.

I think in most cases, the Goldilocks medium one.

Each of these was maybe a tenth of a cent or a fifth of a cent.

And I'm sure a cheap one would be a fifth of that cost and so on.

So, super worth it for what we're getting.

All right.

Here we go.

Let's pop into the perplexity methods folder.

Okay.

That was the formless analysis.

Goals.

Accessible.

Rigorous.

Applicable.

Meremecology.

Notebook.

L.M.

OpenAI API for the translation.

Now, perplexity API.

So, test perplexity API is a short script that tests the network connectivity.

Simple perplexity is where you can paste in your private key.

It's just simpler and keep it all in one just with their example of starting.

So, this is if you want to study the simplest perplexity call you can make.

So, now here we get an LLM keys dot key.

It's in git ignore.

So, it's not sync to GitHub and it has the perplexity API key.

So, there are, as the simple perplexity shows, you can send your message with a system prompt and a user prompt.

So, this is kind of the context or the character system prompt and then the specific user request.

So, first, the user prompts.

This is what somebody would type in.

They're in a JSON called user prompts dot JSON.

And they're numbered as well as given a short name and the full prompt.

So, here prompt one, short name is climate change impact.

And the prompt is conduct a comprehensive meta-analysis of studies published in the last year on the impact of climate change on ant species diversity.

And then extra information.

You could say output in the style or make sure you do a ton of research.

And so, there's several dozen.

The first few dozen are ants related.

Then applications of ants.

What's the role of neurotransmitters, brain regions?

Then prompt 2425, we get into doing meta-analyses of the career of Professor Deborah Gordon and Professor Tim Linksfayer.

We can add more.

If people have any requests for prompts, please write them in the chat and I'll add it here.

Doing meta-analyses on different areas of ants, biometrics, like relatedness within and across colonies, phenotypic correlation measures.

Then, now prompt 33, 34, 35, we're getting into specific ant groups doing a meta-analysis of studies published in the last three years.

So, Camponotus, Formica, Pheidoli, Atta, Solenopsis, Monomorium, Lasius, Oecophila, Pogonomyrmex Barbatus, that's genus and species.

And then non-Barbatus Pogonomyrmex species.

Then we get into the regional.

We got northeast, southwest, USA, Pacific Northwest.

Agricultural ants.

Northern California ants.

Davis.

Davis Agricultural ants.

Davis Urban Ant Ecology.

Davis Invasive Ants.

Ant Symbolism.

Ant Wisdom.

Ants in Contemporary Art.

Ants Indigenous Wisdom Stewardship.

Then 65 takes a bit of a twist.

Here we get more into prompts that are not necessarily related to ants, but some of them do as well.

Relating to the active inference, theoretical foundations, recent developments, applications to ant behavior of active inference.

PhD proposals for active inference.

Foraging simulators for active inference.

Recent events related to active inference.

Writing step-by-step tutorials.

This one's going to be fun to look at.

Giving comprehensive overview of Bayesian mechanics, quantum active inference.

Writing a summary and step-by-step on rxinfer.jl, Julia package.

Writing who's on first dialogues.

Writing grant proposals.

Fire risk assessments.

And county level fire risk assessments.

Thanks, Tucker.

Enjoy.

That's user prompts.

System prompts.

There are fewer, but if someone suggests one, just write it in the live chat.

These are also identified with a number and a short name and a longer description.

So here's a myrmecologist.

Several ant researcher roles.

Programmer.

Complex system scientist.

Local journalist.

Enthusiastic undergrad.

System prompts.

System prompts seven and eight are questions from the innovators catechism at the early ideation and then later pitch stage.

So they're going to respond with question and answer structured information, which we'll look at.

And some other areas.

So then perplexity.py.

You list the short name of the target system prompts and then the target user prompts.

So we could add one if somebody types another user prompt.

So let's do lake county fire risk just to show what it looks like.

Save perplexity.

Run it as python3 perplexity.

So where the outputs already exist,

then it skips it for that combination.

So it looks like that was already done.

Let's see if it'll autocomplete.

Or let's just pick a user prompt that hasn't been done.

Let's do Pheidole Formica recent studies.

So systems,

Innovatory Catechism's pitch.

The outputs existed for some of these

user prompts already.

Then it hits

Formica recent studies,

which is a new one.

So it's going to send that off

to perplexity.

While that's chugging away,

I'm going to go over to the GitHub

and we're going to look at a few of the outputs

of the primary artifacts themselves.

So just the outputs of the perplexity,

different characters doing these different

kinds of syntheses.

And then we will look at the

semantic and linguistic methods

that analyze the folders of documents.

The markdown output folder is nested

into the different roles.

So let's compare

the Innovator's Catechism

with the complex system scientist.

Okay.

Here,

we have complex system scientist

recent events.

Here's IY number four.

This was an event,

I don't know if the dates were correct,
but that did happen in 2023.

There's some
institute quarterly roundtable.

So it's over learning on that
or somehow it came to learn
about that one
live stream.

Here's Bayesian mechanics
overview.

This is written like a
curriculum explainer.

Here's the career meta-analysis
of Professor Deborah Gordon.

Early career topics,
methodologies,
key findings.

This is a different Deborah Gordon.

Both Deborah Gordons,
so it wasn't sure,
so it wrote
about Deborah M. Gordon,
the mymecologist,
and then other Deborah Gordon
as well.

Here's Lake County,
California fire risk.
current fire conditions,
historical fire data,
climate weather pattern,
vegetation,
fuel load,
topography,
human factors,
infrastructural vulnerability,
fire management resources,

prevention mitigation strategies,
future projections,
economic impact,
ecological consideration,
community preparedness,
post-fire recovery,
actionable recommendations,
no citations provided.

Here's overview of RxInfer.

This is the perplexity example
on the web interface
that I showed earlier.

Here it is just in GitHub.

And we can test running this,
but this is pretty much
what it looks like.

And if it's pulling
from the coin toss example,
this should be good.

here's the grants proposal
for safe OSE.

Again,
that was a one
where I showed the
real web interface.

And then let's see
the who's on first.

So this one
makes a who's on first
about Hurricane Milton.

I guess
that's what it thought
current events were.

Now,
if we look over
in the
Innovator's Catechism
system prompt folder,

there should be
some differences here.

Here,
the output format
is requested
to be
the project catechism.

So,
with interesting
made up
theme names,
which we'll
see,
I think,
a few of,
we see the catechism
format
of situation,
problem,
and the entrepreneurial
pitch
catechism
specifically.
situation in terms
of key problems,
user segments,
alternatives,
early adopters.

The mission
and the value
proposition,
potential avenues
of approach
and the resources
that they're
associated with,
advantages,
risks,

feasibilities,
channels,
milestones,
metrics,
cost benefit,
big picture,
admin,
folder,
bibliography.

Here,
it's a pitch
about a project
to study
octave inference
ant behavior
and the first
citation is
a real
paper.

Here we have
Innovator's Catechism
for the
foraging simulation
and it includes
some code
methods.

here's a pitch
for the Bayesian
mechanics.

So,
pitching why
we would
want to have
this Bayesian
mechanics
explainer
on border.

Looking at

these citations,
open up all
four of them,
we have
relevant paper
with the
incorrect citation,
Ramsted et al.,
rather than
Sokativa de Valle
et al.

second paper,
Ramsted et al.,
2023.

Here,
incorrectly as
Royal Society
Publishing 2023.

title is
correct.

Here's
a conference,
2024,
2024,
okay.

Here's
a PDF,
Zuckerman and
George,
Chem Archive.

So,
citations,
not fully
accurate,
but
citations,
real links,
nonetheless.

Here,
we get
to the
grant
proposal.
So,
it's
a
pose
or
safe
host
type
application.
Here,
it
settled on
Linux kernel
open source
ecosystem.
So,
it lays
out,
using the
catechism
format,
the
essential
value
proposition,
avenues of
approach,
all the
information
about the
grant.
Like,
Heilmeier

Catechism
for DARPA,
this is
using the
innovator's
catechism
pitch for
safe host
for NSF
in a
hypothetical
world.

So,
each of
these
folders,
depending on
how the
prompts
interact,
there's
really
interesting
outputs
in terms
of maybe
the
undergrad
making
the
pomdp
pi
mdp
active
inference
simulation,
maybe
this is

the best.

that's
where the
perplexity,
so here
it was
from running
the
perplexity.py,
they all
went for the
formica
recent studies
just while
this was
happening.

so let's
push that
update.
formica
005

!
push it
Now let's
look at
local
journalist
formica.

So this just happened now.

Meta-analysis of recent studies on the genus Formica, 21 to 23.

Here's a table looking pretty nice with two studies on Formica.

Table with one study on social parasitism.

One study on foraging activity, trophobiotic relationships, social parasitism, host parasite.

Different methodologies, significant results and impact, controversies, debates, conclusion, and the formatted references.

Pretty cool. Let's look at the futurist for Formica.

Here it's a different format.

We see more of a general summary of the state of the art on the monogynepolygyne research agenda.

We don't see those tables broken out the same way.

And there's not a strong futurist bent.

So it could be some interaction of the model, the system prompt, the user prompt.

Of course, all this needs to be checked and could be checked with synthetic as well as manual human means.

Let's see what the myrmecologist...

What would the myrmecologist say when it walked into a NSF grant application?

So it wasn't included because it was taken off these targets.

So here, add in comprehensive.

Let's just check from system prompts.

It's called comprehensive myrmecology.

Paste it in.

Save it.

Clear it.

Run perplexity.

Oh.

Maybe there already is...

That.

Okay.

So that's the combinatorics of generating combinations of the user and the system prompt.

This is going to now get into the visualization section that's new from the report.

Visualization methods dot py is a nice long...

590 line...

...methods file.

Visualization dot py brings in those methods.

Set the input and the output folder.

Make the directory for the output if it's not there.

Set up the logging.

Load in...

...some of the text processing modules.

Pre-process.

Perform some of the analyses.

Output several groups of visualizations.

Group one is related to dimension reduction visualizations.

Group two is related to enhanced PCA.

Principal component analysis visualization.

That was part of group one.

Group two, word importance and PCA.

Group three is more text analysis.

So...

I'm going to delete the folder visualizations.

Right click.

Open integrated kernel.

Python 3.

Visualization.

So this is what it's going to look like when it runs.

It's going to output how many markdown files it's taken in.

158.

So if we added more to perplexity dot py, we could do some of that.

We could add another.

And then it would output more.

And then rerunning visualization would pick up on those new ones.

Okay.

So there's nothing in visualizations yet.

It's doing the analyses.

And then it's going to start to spit one by one saved images into the visualizations folder.

We will look at those visualizations.

And see what it shows us about these outputs in terms of their semantics.

And in terms of how the system prompt and the user prompt interact.

And how we might use that.

So the analysis gets slower as more documents come into play.

I'll just show.

Here's system prompts dot JSON.

Control A.

Control I.

So bring it in the composer.

Add.

A.

System prompt.

For.

An expert.

Chef.

Who outputs.

Dinner.

Menus.

With.

Rich.

Relevance.

For any.

Prompt.

Output.

Output is always.

Epic.

Dinner.

Adventures.

Though.

So this is a way to use cursor.

In this informal way.

To.

Write a well structured prompt.

13.

So it's going to first write it in this side window.

That's kind of.

And then.

It scans over.

And now.

Culinary adventurer.

So.

Take culinary adventure.

Go back to perplexity.

Systems prompt.

Culinary adventure.

It.

I think that's super maven.

But still.

It knew.

It knew.

That I wanted that one.

I'm going to.

Terminate the visualization.

Effort.

For now.

Call the.

Python 3.

Perplexity.

So now here's the new.

The ones that were already done.

Are skipped.

And the new.

Combinations.

Are being.

Sent off.

Right click.

Open the integrated terminal.

Python 3.

Visualization.

Other outputs.

Was just.

Earlier.

Texts.

From.

From a simpler time.

From a different time.

I'll.

I'll delete these.

In.

They're.

They're always there on git.

But I'll delete them.

For now.

Push to git.

Update.

And then.

Visualizations.

Are going to start.

To come through.

And this is going to.

Help us get.

A handle.

I'm also going to.

Resync the cursor.

Context.

Up.

It was.

It.

Resynced.

Great.

Okay.

Introduce.

Perplexity.py.

Explain the structure.

Of user.

And system.

Prompts.

Looked at.

Simple and test.

Explaining how different kinds of prompts are handled.

Let's look at the visualizations.

And then we'll go to the next.

Steps.

So again.

Here's visualization.

Dot.

Py.

All right.

All right.

So.

They're now getting.

Output to the folder.

I'll just let them all.

Run.

Let's just start with.

LSA.

Plot.

So here's two components.

Looks like.

Partially labeled.

LSA component one.

Document semantics.

PCA 3D plot.

Okay.

Cool.

Interesting.

Color scheme.

PCA.

PCA.

Cumulative variance.

So it looks like only five PCA components are retained.

We can increase that number.

But.

It's showing the accumulation of variance.

And we're not.

Saturated yet.

So.

To capture more of the variance.

It would probably make sense to add more than five principal component.

Terms.

Okay.

Confidence intervals.

This is the mean.

TF IDF.

Score.

Which is a document frequency.

Inverse document.

Frequency.

So here we have.

Ant.

And inference.

With higher scores.

And then some of these other ones.

With lower scores.

LSA topics.

Okay.

Here we have different ways of doing topic.

Modeling.

Topic one has to do with time passing on graphs.

And reactive message passing.

In the Julia package.

As well as Luke Costello.

For some reason.

From the who's on first probably.

Topic two.

Having to do with.

Active inference ontology terms.

Quantum active.

Inference.

Topic three.

Having to do with collective behavior.

Free energy.

Four.

Having to do with the.

The other Deborah Gordon.

Talking about the environmental.

The world.

Resources modeling.

Here's links there.

And security.

So.

Okay.

Cool.

Here's the.

Principle component analysis.

With the eigen terms.

So.

We looked at the cumulative variance plot before.

The screen plot is showing the decreasing fraction of variance that each of these.

Components add.

Let's.

Let's.

Let's update.

Visualization.

Methods.

Ensure to include.

50 PCA.

Dimensions.

Okay.

We'll keep looking at it.

There's the principal component analysis.

Oh.

We were.

We were high up.

This is pretty cool.

Here we have.

Five of the different.

Principal components with colors.

And then.

We can see.

Okay.

Fire.

Is.

Cross listed.

On.

Explaining the variance.

Fractionation.

Of.

Blue and green.

Whereas.

Inference.

Is only loaded up.

On.

Fraction two.

And then.

That's shown here.

They're separate.

But also.

There's some.

That are cross listed.

Let's see.

How that edit.

Went.

Okay.

Interesting.

Looks like it's deleting.

A bunch of stuff.

Let's accept it.

Save it.

Clear.

Run the visualization again.

So now.

These.

Culinary adventurers.

Are still pumping.

In the background.

So.

Now.

We have.

169.

Markdown files.

Whereas.

In the previous time.

There was.

158.

Or something like that.

So it's going to.

Redo.

These.

They might be changing.

Right from underneath us.

Okay.

Prompt distribution.

Here's.

Different system prompts.

And.

And.

User prompts.

And different words in them.

Here's a funny.

Visualization.

Document length.

Across the system prompts.

So here we can say.

Okay.

The expert programmer.

Has a shorter.

Document length.

On average.

And then here's.

The statistical tests.

And the.

The p values.

For the ANOVA.

Alpha.

It's not statistically significant.

At a.

0.05 alpha.

Term frequency.

Here's.

Term frequency.

Inference.

Ant.

Fire.

Term network.

So here we have.

Terms.

And their.

Connections.

With the.

Co-occurrence.

Module.

So this is a lot.

Like gene.

Co-occurrence.

Based upon.

Expression levels.

Here's.

Term frequency.

Heat map.

So we can see.

Red is.

More usage.

More unique.

Strong usage.

Light yellow.

Is.

Less strong.

Unique usage.

Show.

Fire.

Is a word.

That's either.

Mentioned.

A lot.

Or it's not.

There's other terms.

Such as.

System.

Or research.

Or use.

That have more.

Variable.

Use.

T-SNE plot.

Word cloud.

And.

Word importance.

Word importance.

Here's the top 20 words.

In each of those modules.

And then a word cloud for them.

And similar for PCA.

And for the LSA.

This most recent.

The.

Culinary adventurer.

So.

There they are.

Culinary adventurer.

That's just showing how.

You can.

Add.

To the system prompts.

Bring it into perplexity.py.

Have it combinatorically generating.

Using the perplexity API call.

So.

So.

Then we'll.

And then.

Meanwhile.

You can be rerunning.

The visualization method.

So here there's an error.

Fix the error.

To fix the error.

Image size of 9000 by 90,000 pixels.

Is too large.

It must be less than 2 to the 16th.

In each direction.

Whoa.

Because we asked about.

Retaining 50 dimensions.

Save it.

Clear it.

Delete the visualizations folder.

Run it again.

We could parallelize.

The API call.

Get closer to that rate limit.

50 per minute.

Let's look at some of the outputs.

Reload.

Markdown output.

Culinary adventurer.

That's the new one.

So we asked.

This is what is so fascinating.

This is really worth the ponder.

Who's on first of current events?

I've seen this several times.

This topic.

Wow.

Look at the topics here.

Vaccine.

Elon Musk.

Artificial intelligence.

Hurricanes.

Meta.

Mark Zuckerberg.

Putin.

Vladimir Putin.

Google.

And it's all unpacked.

Unpacked.

That's not a recipe.

That's not a menu.

Fire assessment.

Not a menu.

Formica.

The prompt did work.

So here.

The societal symphony of formica ants.

A dinner concept.

Colony awakening.

First dish.

Appetizer.

Forager's bounty.

Super salad.

Super colony soup.

Main course.

Queen's feast.

Palate cleanser.

Aphids nectar.

Wow.

Wow.

Chef's commentary.

With references.

Analysis of active inference events.

Same.

Outputs as others.

Okay.

Let's check back.

Looks like the PCA.

Let's check back at the PCA.

Here we go.

Cool.

So now we have.

Cumulative variance.

Rarefaction.

Plot.

And we can see a lot more richly.

The relationship with these different terms.

Loading's heat map.

Here's a thousand features retained.

You can see some patterns of positive and negative.

Here.

We see different PCA plotting.

Different PCA plotting.

There's some methods that look at where things start to bottom out or change.

Like break points.

Here's the terms that are associated with the top 50 principal components.

Term network.

So that's pretty cool.

And.

Let's add another user prompt.

Write.

Write.

With code.

A workshop.

A hands-on.

Workshop.

Oriented towards.

Applying active inference.

In the bioregional.

Setting.

Add a prompt.

That will.

Write.

With code.

Comprehensive.

Hands-on workshop.

Oriented towards applying active inference in bioregional settings.

Be.

Professional.

[illegible]

So prompt 81.

So prompt 81.

Move that over there.

Rerun perplexity.

Here we go.

So skips over what's already been done.

First one that it hits is Innovator's Catechism Pitch, the first one in the list.

Catching on the first one that is undone.

So once that first one comes out, I'll push it.

We'll check it out on GitHub.

It'll be the value proposition from an Innovator's Catechism Pitch perspective for the Bioregional Active Inference Workshop.

41 seconds.

Okay.

Push it.

Innovator's Catechism Pitch.

It's uploading all the images.

Reload it.

Bioregional AI Workshop at gmail.com

Situation.

Key problems.

Lack of integrated decision-making tools.

Innovation.

Resource management.

Limited collaboration.

User segments.

Ecologists.

Environmental managers.

Data scientists.

Alternative.

Traditional statistical models, which are static and often static and do not account for dynamic environmental changes, machine learning approaches.

While powerful, these may not provide the same level of interpretability and integration with ecological principles as active inference.

Early adopters.

Research institutions.

Government agencies.

NGOs.

Community groups.

Mission.

Value proposition.

To provide a comprehensive and practical understanding of active inference principles and their application to bioregional management, enabling participants to make more informed, sustainable, and adaptive decisions.

Avenues of approach.

Theoretical foundation.

Mathematical framework.

Computational implementation.

Data collection and processing.

Pre-processing.

Model building and simulation.

Case studies.

Integration with existing management practices.

Future directions.

Advanced topics.

Resources.

Computational resources.

Data and human resources.

Advantages.

Unified framework and interpretability.

Risks.

Complexity and data quality.

Feasibility.

Given the advantages and the growing interest in active inference, this project is highly feasible.

The workshop can leverage existing research and tools and the facilitators can provide the necessary support to ensure participants understand and apply the concepts effectively.

Channels.

Academic and professional networks.

Online platforms.

Milestones.

Metrics.

Milestones.

Cost and benefit.

Big picture.

Big picture.

Dr. Maria Rafa.

Day 1.

Theoretical foundations.

Mathematical framework.

Morning session.

Afternoon session.

Day 2.

Computational implementation and data collection.

Day 3.

Model building.

Simulation.

Case study.

Day 4.

Integration.

Future directions.

Advanced topics.

Follow-up exercises.

Wow.

Wow.

Expert programmer.

Ants.

Ants, contemporary art.

Ants, geopolitics and social structure.

Expert programmer, ant foraging.

Local journalist, New Mexico fire risk. Looks a lot like the other ones.

Okay.

Okay.

So that is basically all.

This has been a checkpoint for the form index project.

There's some Ant specific directions. There's some science, education, communication, generative AI, plus plus plus.

A lot of fun and interesting methods that people may find useful.

where are we scouting or playing or researching on the edge and beyond

where we can change something like a cul-de-sac and it won't influence other things.

Get the feedback from Formus.

Improve the methods.

Integrate with Species ID.

Do perplexity searches for all species groups of ants and maybe beyond.

Integrate with Fabric.

There's no written questions yet.

If anyone has ideas, leave a comment.

Get in touch.

I hope this was of epistemic and pragmatic value for you.

Check out the repo.

Check it all out.

End of number 005.1.

And...

There it is.

End of number 005.1.

End of number 005.1.

All right.

And there's the Zenodo.3 version from the beginning of the stream.

Pretty cool.

Awesome times.

Okay.

Thanks for watching.

Looking forward to people's feedback where it all goes.

Till next time.

Bye.

Bye.